THE PUSHFORWARD OF AN INJECTIVE MODULE IS TOTALLY ACYCLIC

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the criterion applied is the companion lemma of the same section and the adjunction computation is carried out inside the counted span.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Totally acyclic sheaf and the criterion characterising it. Same chapter, the definition of total acyclicity and the criterion by the exactness of extended Cech complexes, which is the companion lemma of this section. Supplies the property verified for the pushforward of an injective module and the criterion used to verify it.
- Injective object of Mod(O_C) and the adjunction between pullback and pushforward. Same chapter, injectivity of a module over a ringed topos and the adjunction f^{-1} – f_* for modules. Supplies the exactness of the functors used to transfer the vanishing of cohomology to the pushforward.

Source reference. Original lemma 03CR: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 2224–2224. Navigation pointer invoked at proof line 2224; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03FC: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 1816–1816. Navigation pointer invoked at proof line 2243; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Source reference. Original lemma 03F9: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 1565–1565. Navigation pointer invoked at proof line 2246; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $f : (Sh(\mathcal{C}), \mathcal{O}_\mathcal{C}) \to (Sh(\mathcal{D}), \mathcal{O}_\mathcal{D})$ be a morphism of ringed topoi. Then for any injective object $\mathcal{I}$ in $Mod(\mathcal{O}_\mathcal{C})$ the pushforward $f_*\mathcal{I}$ is totally acyclic.*

Proof. Let K be a sheaf of sets on $\mathcal{D}$. By Modules on Sites, Lemma 03CR we may replace $\mathcal{C}$, $\mathcal{D}$ by "larger" sites such that f comes from a morphism of ringed sites induced by a continuous functor $u : \mathcal{D} \to \mathcal{C}$ such that $K = h_V$ for some object V of $\mathcal{D}$.

Thus we have to show that $H^q(V, f_*\mathcal{I})$ is zero for $q > 0$ and all objects V of $\mathcal{D}$ when f is given by a morphism of ringed sites. Let $\mathcal{V} = \{V_j \to V\}$ be any covering of $\mathcal{D}$. Since u is continuous we see that $\mathcal{U} = \{u(V_j) \to u(V)\}$ is a covering of $\mathcal{C}$. Then we have an equality of Čech complexes

$$\check{\mathcal{C}}^\bullet(\mathcal{V}, f_*\mathcal{I}) = \check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{I})$$

by the definition of f_*. By Lemma 03FC we see that the cohomology of this complex is zero in positive degrees. We win by Lemma 03F9. □